

Prediction of bank customer churn

By

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Project Business intelligence

Customer relationship management (CRM)

- Acquisition
- Creating value for customers
- Retention

Supervised machine learning

- Regression
- Classification

Customer relationship management (CRM) is an integrated strategy for locating, attracting, and keeping customers (Buttle and Maklan, 2019). They reported that it covers every facet of interacting with customers and business prospects in order to enhance long-term growth with client and company profit.

Among the CRM approaches are customer acquisition, creating value for customers, and customer retention (Bolton & Tarasi, 2017).

Supervised machine learning models the relationship between the independent (explanatory) variables and the dependent (outcome) variable (Valletta *et al.*, 2017). They explained that this learning occurs in regression and classification.

Data generation and mining tool

Public data

- Kaggle

A publicly available dataset that is applicable to this business intelligence project was retrieved from Kaggle (Radheshyam, 2023) and mined to solve the task. Below is the link to the dataset.

<https://www.kaggle.com/datasets/radheshyamkollipara/bank-customer-churn/data>

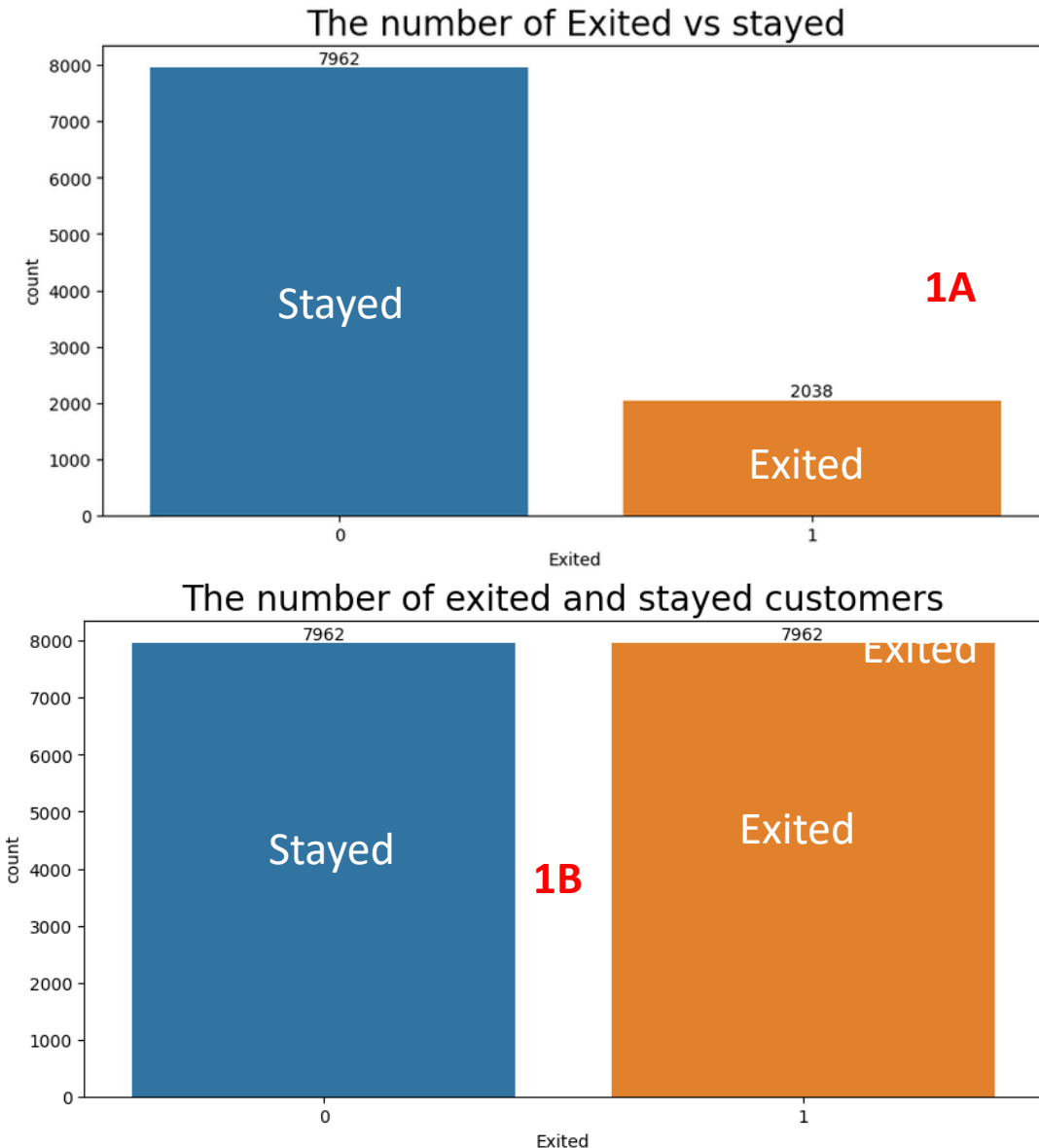
Data mining tool

- Python programming language

Extensive numbers of Python-based machine learning libraries with several algorithms for efficient modeling are highly distinguishable and have made the Python programming language popularly used (Nabeel, 2022).

The detailed codes for this data modelling-based business intelligence task can be accessed in my GitHub account via this link: <https://github.com/Latmos-G/BI-portfolio/blob/main/BI%20project.ipynb>

Exploratory data analysis

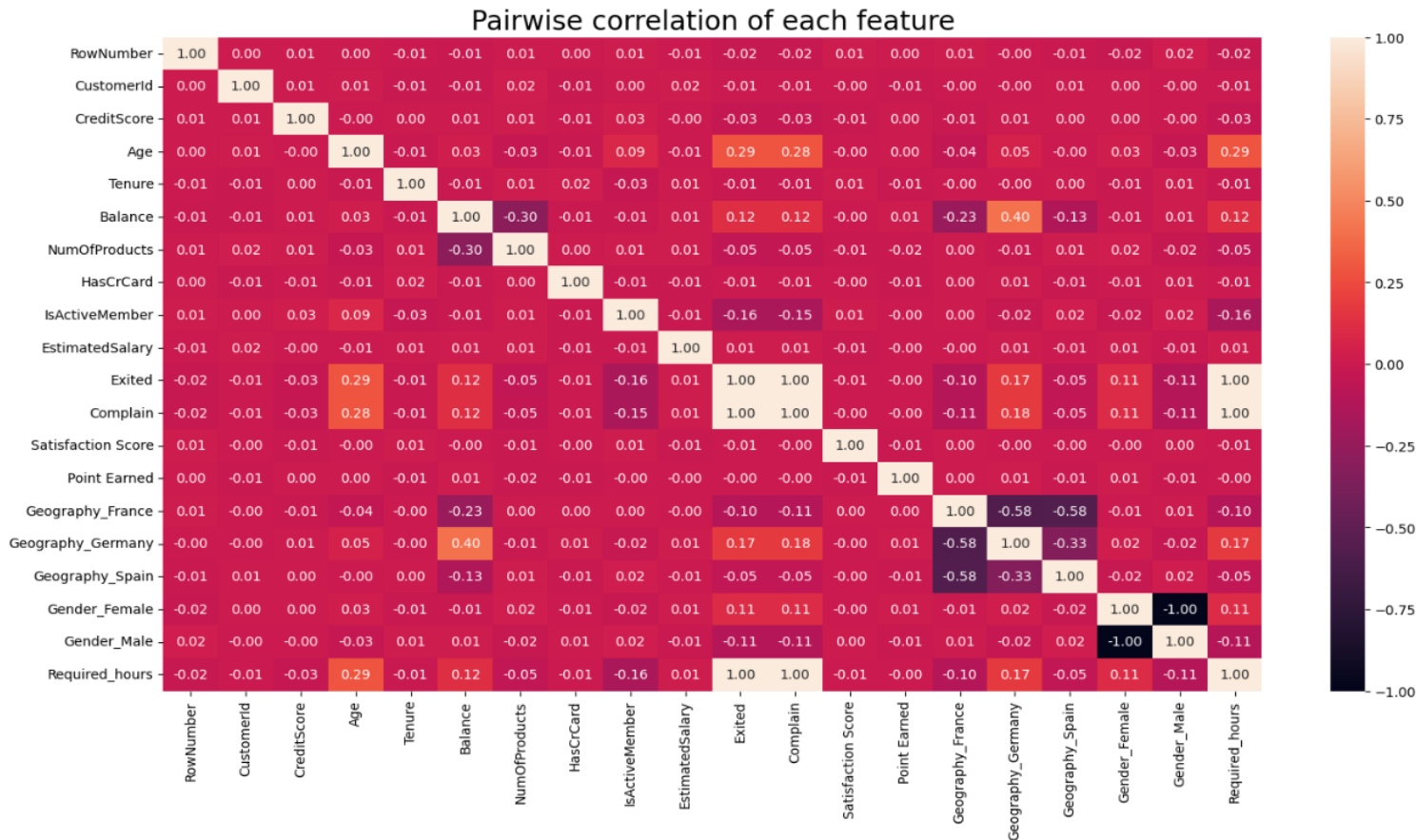


This chart represents the counts of exited and non-excited customers.

The count frequency in the raw data (1A) and the randomly balanced counts (1B) by oversampling technique.

Balancing the imbalanced datasets prevents the machine learning algorithms from learning biased features. (Ashraf, 2023)

Figure 1. shows the counts of the outcome variable



The pairwise correlation for all variables (Figure 2) is displayed here.

The closer the values are to 1 and -1, the stronger the assumed relationship between the concerned two variables.

This will guide the management on some variables that correlate to customer exit or not.

Figure 2. shows the pairwise correlation of the entire features.

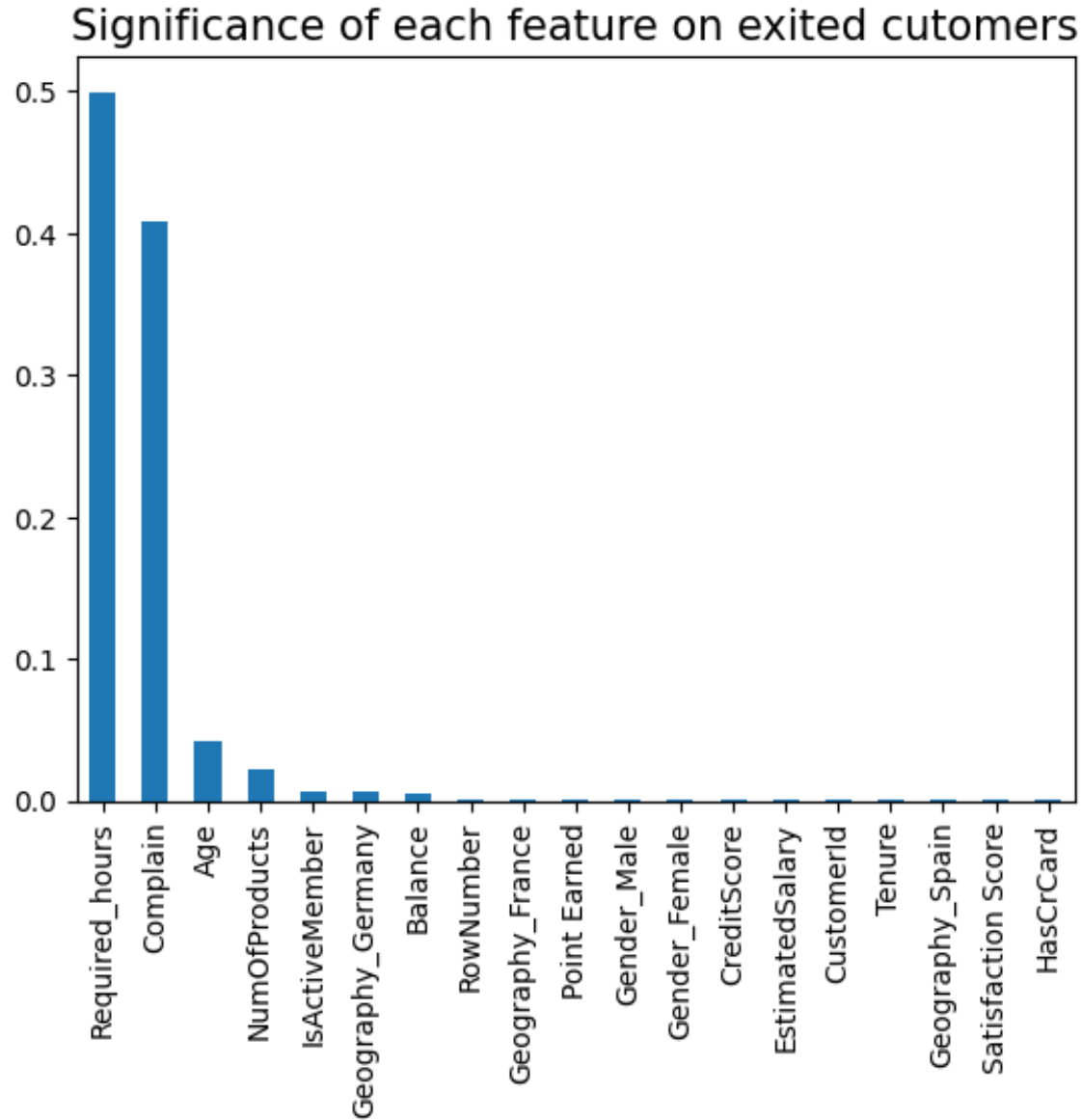


Figure 3 shows the important features for the classification task

The important features to be considered for the predictive modelling (Figure 3).

This will prevent using non-essential features that can introduce spurious correlation in the predictive modelling (Ye *et al.*, 2024). Required hours, complain, age, number of products, and active member are the most important to the churning of the customers.

An accurate model is the prerequisite to guide an accurate decision to be taken by the management.

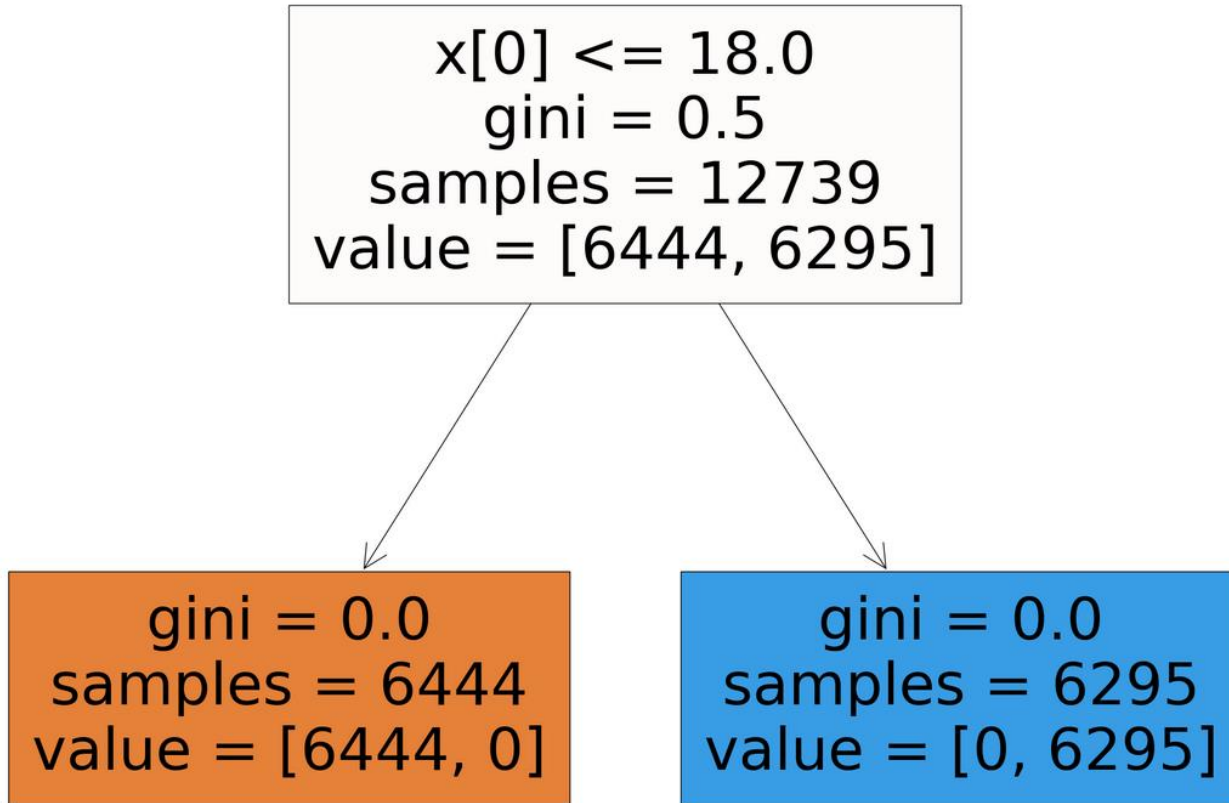


Figure 4 shows the grouping achieved by the decision tree on the turn-around time and the churning of customers.

How can the product process be improved within less time?

The decision tree algorithm (Figure 4). has been reported for classifying binary outcomes of supervised learning (Valletta *et al.*, 2017).

This emphasizes its suitability to model the effect of the 'required hours' to meet the customer needs and categorize customers that exited and non-excited.

Exited of 6295 customers amount to 49.4% of the total estimated customers due to turn around time, this is a huge loss to management. Addressing the turnaround time shows to be one of the means towards customer retention.

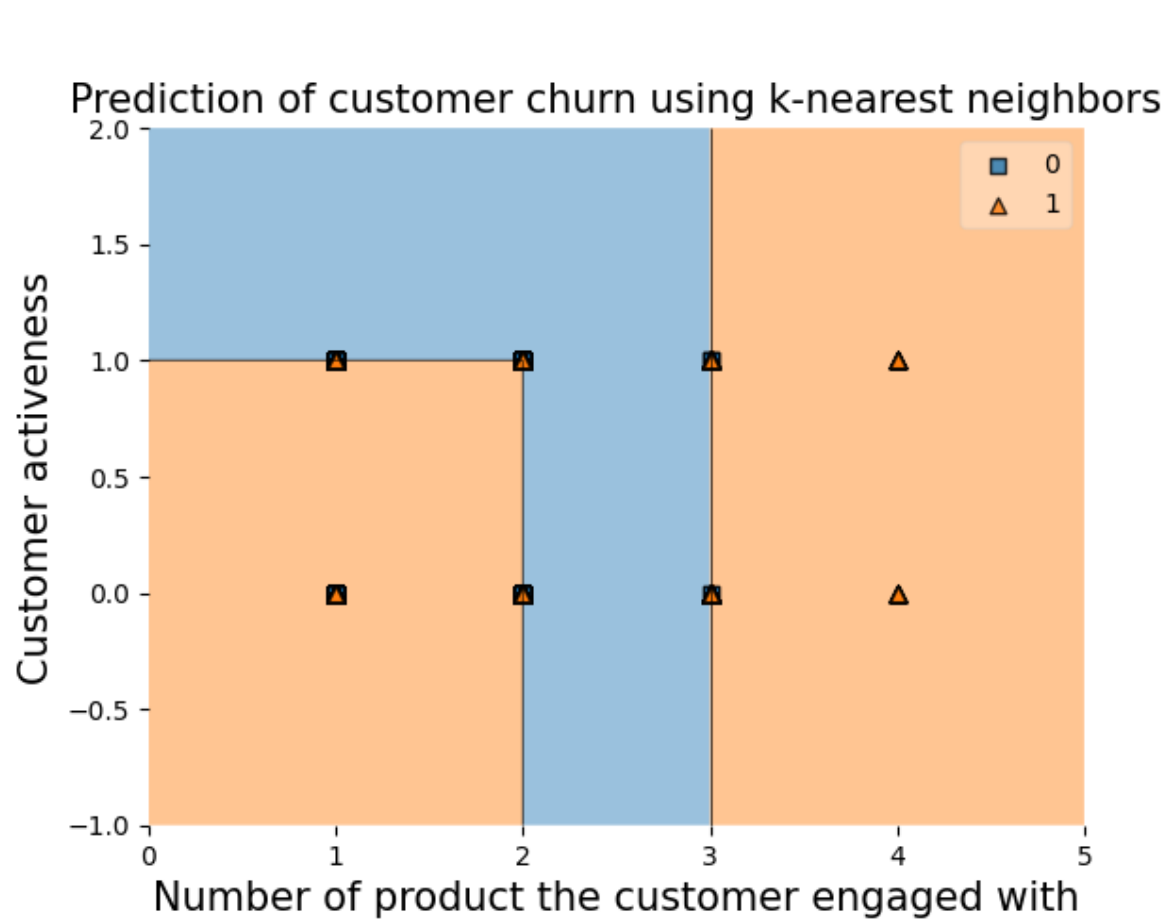


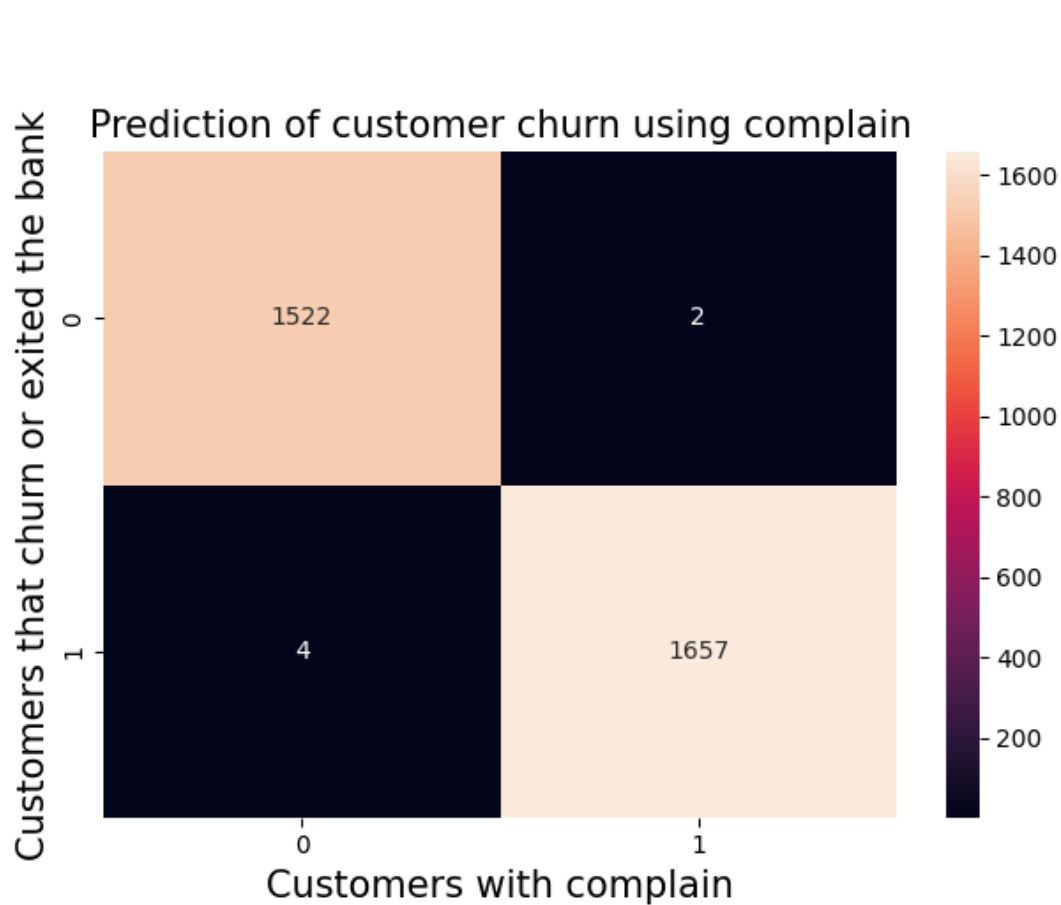
Figure 5 shows the grouping achieved by the K-nearest neighbors on the number of products and their activeness in using the product.

Which equity
void of bias is to
be employed?

The K-nearest neighbors (KNN) algorithm (Figure 5) is a supervised learning algorithm for this classification task (Nabeel, 2022).

It is justified to model the number of products' and the customer's activeness to the own product' on the exited customer.

The management needs to look for incentives per product owned by customers. This will ensure equity gains to them respective to their products, as the below observations reveal from the KNN algorithm. The customers with 2 to 3 products mostly stayed while customers with 4 products exited the bank irrespective of the customer's activeness.



What plans are appropriate to prevent the defection of non-satisfied customers?

Logistic regression was evaluated with a confusion matrix (Figure 4), a binary classifier algorithm of supervised learning (Nabeel, 2022).

Its suitability triggers its choice to model complaints with exited customers.

Exited of 1657 customers amount to 99.9% of the total customers that complain. This is a serious red flag to the management that any customer that tender complains is on the verge of exit. A measure must be taken to review the nature of the complaints and prevent their re-occurrence.

Figure 6 shows an evaluation of the logistic regression of modeling complaints to classifying customers.

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